Science · LAUNCH · Autumn 2 · Week 1 · SCI-LAUNCH-A2-W01-L3

Guidance and staff card

At check-out, record E for … against the pupil's target.

At check-out, record E for justifying the cell-cycle order and explaining how matching chromosome sets support growth and repair against the pupil's target.

Prepared 3 November 2026

Invite a response. Give thinking time. The learner may pass. Offer private answers and the usual response mode.

Return to learning: use the individual pause plan, give space, then return to the last secure step and one structured choice. Correct privately.

Use soft instrumental music for transitions if the class plan says so.

Code	Meaning
VF	Verbal feedback given here
WS	Worked with support
I	Independent
NS + …	Next step one specific thing
E	Evidence captured on the evidence app
R	Responded loop closed after a pupil response and adult engagement
//	Self-edit point
?	Read this back to me

At check-out, record E for today's lesson against each pupil's target on the evidence app

Use observed evidence and record the support given. An edit is in the same colour, on the same page. Written feedback is used for extended writing when it adds value.

Codes follow the verified feedback policy, Issue 1. A button, saved answer or completed stage does not award a code.

Feedback codes VF · WS · I · NS + … · E · R · // · ?
Give one feedback code. My response is the pupil’s improvement to the work.
Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt

Teacher and cover plan

At check-out, record E for justifying the cell-cycle order and explaining how matching chromosome sets support growth and repair against the pupil's target.

I am learning to justify the cell-cycle order and explain how matching chromosome sets support growth and repair.

Source sequence retained: Arrival 5; Starter 3; modelling 5 split across the two I do stages; first We do 7; second We do 5; You do 8; Review 4; Exit 3. Total 40 minutes.

Week arc: Discover: the cell cycle. Use: mitosis in sequence. Master: matching chromosome sets and repair.

Kit: large prepared paper chromosome strips, two cell outlines and a tray. Use the live lesson’s original copy, separate and split model, full and faded sequences, number hinge and two-fault challenge.

Keep the three settled reset opportunities inside shared work, independent work and review. Use an agreed pause and one-step return; add no separate stage.

Teaching sequence

Stage	Minutes	Teaching and check
Arrival	5	Q1 checks picture access. Compare the first and last pictures. Offer enlarged or tactile paper shapes; accept pointing before speaking.
Starter	3	Ask the original three retrieval questions in order. Invite a response and allow a pass. Reveal answers separately and revisit the success criteria.
I do	3	Show the complete original model. Step or play the six frames, then pause: is DNA copied before, during or after mitosis? Answer: before, in interphase.
We do	7	Use two quiet partner turns before writing. Build the six-stage order and justify copying, separation or the final split. Check the first out-of-place stage, then hear a reason in the learner’s usual mode. Use a fresh stage choice to check independence. Respond to the optional participation request by making or explaining a change and checking its effect.
I do	2	Model the six-chromosome case. Separate nuclear division from the final split. Ask what each new nucleus must receive.

Stage	Minutes	Teaching and check
We do	5	Collect simultaneous whiteboard, symbol or pointing responses and hear separate reasons. About three in four secure: move on. Otherwise re-model with two blue and two orange paper strips, then use the eight-chromosome parallel hinge. Correct answer: eight. Do not let a group result replace an individual response.
You do	8	Use the original order, matching-set, growth and repair content to hear what each learner can explain. Keep the two-fault challenge available; Choice offers selected answers plus a reason. Give access support without dictating the science. Record actual independence and prompt level.
Review and improve	4	Give one specific feedback point. Wait for a real pupil edit or explanation and engage with it. Record only feedback that occurred.
Exit ticket	3	Hear every response through the usual mode. Use it to change the next example or prompt. Keep one response and state the support actually used.

Stage timings

Stage	Minutes
Today I can	0
Arrival	5
Starter	3
I do	3
We do	7
I do	2
We do	5
You do	8
Review and improve	4
Exit ticket	3
Tools & print	0

Total: 40 minutes

Feedback codes VF · WS · I · NS + … · E · R · // · ?
Give one feedback code. My response is the pupil's improvement to the work.
Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Exit pupil voice
What helped you most today, or what would you change next time?
Accept a pointed, spoken, signed, drawn, written or chosen response. A pupil may pass.

Made by Matt

P17 · SLT planning and evaluation

Planned adaptations

Need	Planned adaptation
Cognition and learning	Start with copy, separate and split, then place the four mitosis phases. Keep the full reference available and fade one phase cue after a secure justification.
Communication and interaction	Two quiet partner turns and the same speaking frame before writing. Accept point, sign or an exact scribe.
Regulation	Check access before the hinge and independent task. Use the agreed pause and one-step return privately.
Sensory and physical	Enlarged, high-contrast, easy-grip chromosome strips. Offer seated or standing handling and adult-directed placement.
Language access	Pre-teach the science terms with diagram and example. Allow planning in any language at the same scientific challenge.

If behind

Preserve copy, separate and split, each learner’s matching-set explanation, review and the two exit checks. After a secure hinge, shorten repeated oral rehearsal. Record the change and coverage impact. Use the next lesson, Discover: growth and stem cells, to revisit an insecure phase cue.

Where catch-up is embedded

Arrival begins with the same labelled picture. The full sequence is available beside the faded one. Reference A and B stay beside You do. The catch-up sheet replaces a starting step after absence or an access barrier only.

Staff evaluation

Prompt	Record after teaching
Delivery	Delivered as planned, partly or changed; actual reason.
Criteria	Criterion, response and support given.
Adaptation	Actual adjustment and effect on a fresh response.
Coverage	What remains and the next teaching move.
Evidence	Location on the evidence app and this lesson code.

P18 · Individual target opportunities

Use opportunities from each learner’s existing plan. Keep pupil codes and target fields blank.

Opportunity	Lesson evidence and adult response
Cognition and learning	Order a fresh phase choice and give one independent scientific justification. Record the pupil’s own sequencing, attention or independence target and actual prompt level.
Communication	Respond to the six-chromosome hinge in the agreed mode. Hear the learner’s meaning without requiring speech.
Regulation	Use the predictable picture start or agreed pause and re-entry step. Evidence the pupil’s own target privately.
Sensory or physical access	Direct or handle the model pieces using agreed access equipment. Separate access from science content support.
Language access	Use the actual vocabulary in the example; allow first-language planning and oral rehearsal. Record an agreed language target where relevant.

One target-matched evidence record can serve the existing plans. Task completion alone does not prove a target. Follow the individual support plan where it differs from these routines.

Made by Matt

Adult deployment and responsive feedback

Agree who supports the independent stage and who holds the group. With two adults, swap roles as useful. With one adult, use a prepared reference task while hearing learners separately.

Prompt least first: wait; point to the last complete step; ask what happened to the copied chromosomes; model one move with one type; let the learner try a fresh move with the other type. Use no physical prompting.

At I do: “Watch the DNA copy. Point to where the two matching sets go.” At We do: “Why must this step come before the next one? Show what each new nucleus receives.” At You do: “Show me the last step you know. What does the model tell you next?”

Record the actual scientific response and content prompt, separately from the access aid. At handover, share what was independent, the feedback and the next step. Do not substitute a group product for an individual response.

Within the optional participation prompt, the lesson teacher hears the request, says what can change, makes one agreed change if possible, explains a limit and checks the effect.

Made by Matt

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First lesson back · once-only read

Before the lesson · scan the books

Scan for five minutes. Find where teaching stopped, existing feedback and actual pupil responses. Look for access barriers and useful science evidence.

Existing NS + is history

Do not chase, back-mark or retrospectively close earlier feedback. Begin fresh feedback from this lesson.

Ask the cover teacher and TAs

Ask what was taught, which model was used, which phase cues were faded and what changed for individual learners.

Check the roll

Compare attendance with the available work. New arrivals start from what they show here; absence creates no automatic deficit label.

Arrival read-off · decide by about minute eight

Read picture access before judging science. If number and matching sets are insecure, use large labelled outlines. If secure, fade a phase cue and ask for a reason.

After last lesson · one read, once

Read each criterion against the actual response and support given. Choose the next example or route from the least secure dependency. Then teach; do not reopen historical feedback.

Made by Matt

Preparation and practical safety

This is optional model rehearsal, not claimed core practical completion. Use large prepared paper chromosome strips, two cell outlines and a tray.

Loose strips could fall or be mouthed. Keep them on a desk or tray, out of mouths, and collect dropped pieces. Confirm the exact kit, local risk assessment and individual plans before use.

Offer enlarged print, tactile paper shapes, a static sequence, high contrast and adult-directed handling. Test pause, reset, keyboard controls and the offline display. Provide the same model on paper.

Destination link: a laboratory technician labels a cell sample and records what its microscope image shows. Keep model symbols distinct from microscope observations.

Made by Matt

Teacher answers and topic guidance

Accept scientifically equivalent answers in the learner's usual mode. Award each distinct point once. Keep answers hidden during independent attempts.

Starter

1. Active transport.
2. Water.
3. The nucleus.

Full and faded sequence

Interphase; prophase; metaphase; anaphase; telophase; cytokinesis. The faded step is anaphase: the two copies of each chromosome are pulled to opposite ends. DNA is copied before mitosis, in interphase.

Order and justify

Interphase, prophase, metaphase, anaphase, telophase, cytokinesis. The four middle phases form mitosis. Copy first: the copies must exist before separation. Separate: one copy of each chromosome goes to each new nucleus. Split last: the final split produces two separate new body cells. The builder previews each chosen stage, checks the first out-of-place position and checks the selected scientific reason. Reveal order replaces the arrangement with the scientific order; Reset restores the starting mixed sequence and clears the reason.

Two faulty models

Fault 1: copy DNA in interphase before mitosis. Fault 2: in anaphase send one copy of every chromosome to each end. A later correct cell split cannot repair missing or unevenly separated chromosome copies.

Six-chromosome hinge

C: six in each daughter, with the same set as the original body cell. A, three: confuses this ordinary body-cell model with chromosome halving. B, twelve: treats copied material as the number retained by each daughter. D, none: leaves the copies in one cell instead of distributing them.

About three in four secure responses: move on. Otherwise re-model with two blue and two orange paper strips, labelled A and B, then use the parallel case. A starting set of eight gives eight in each daughter.

Transfer to repair

DNA is copied before matching copies separate into two nuclei. After the cell splits, matching new body cells can replace damaged cells. Accept the same explanation through a model demonstration.

Science scope and misconceptions

Mitosis separates copied chromosomes into two nuclei. Cytokinesis splits the cell. The model concerns ordinary body-cell division. It does not teach the full molecular detail, asexual reproduction or chromosome halving. Do not describe mitosis alone as the whole final split.

Copy before separation: both new nuclei need a matching set. Uneven separation can leave one nucleus with an extra copy and another lacking it. New matching body cells support growth and replacement of damaged cells.

Arrival Supported

1. Two cells.
2. The one cell on the left.
3. The inner outline labelled nucleus.
4. One cell has become two; each new cell has a matching labelled set.

Arrival Independent

1. Two.
2. The nucleus.
3. Both cells have A and B: the sets match.
4. One cell became two, each with a matching chromosome set.

Arrival Aspire

1. Two separate cells.
2. The nucleus.
3. Each new cell has both types, A and B, so the sets match.
4. It shows one cell becoming two with matching sets; it does not show the full molecular detail or every real chromosome.

Exit Supported

1. So matching copies exist for both new nuclei.
2. They can replace damaged cells during repair.

Exit Independent

1. DNA is copied before mitosis, and one copy of every chromosome separates into each new nucleus before the cell splits.
2. More new body cells support growth; new body cells also replace damaged cells during repair.

Exit Aspire

1. No. Splitting the cell does not redistribute chromosome copies; one new nucleus can gain a copy while another lacks it.
2. Copy DNA in interphase before mitosis. The copies must exist before they can separate into matching sets.

Made by Matt

Mark scheme Supported

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Order interphase, mitosis and cytokinesis. Say why copying must come first. [3]

Interphase before mitosis; cytokinesis last; copying first supplies copies for both new nuclei.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Complete: DNA is ___; matching copies ___; each new nucleus has a matching ___. [3]

DNA is copied; matching copies separate; each new nucleus has a matching set.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Damaged skin needs new body cells. Use copy, separate and split to explain how new matching cells help. [3]

Copy DNA before matching copies separate into two nuclei; split the cell to form matching new body cells; these can replace damaged cells.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Mark scheme Independent

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Order the six named stages. Mark the four mitosis phases and justify why interphase comes first. [4]

Interphase first and cytokinesis last; prophase, metaphase, anaphase, telophase in order; those four identified as mitosis; DNA copying in interphase supplies matching copies before separation.

Maximum 4 marks. Accept equivalent valid wording or a clear model demonstration.

2. Explain how DNA copying and chromosome separation give each daughter cell a matching set. [3]

DNA is copied before mitosis; one copy of each chromosome goes to each new nucleus; after the final split both daughter cells have matching sets.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Explain how this cell-cycle sequence supports growth and the repair of damaged skin. [3]

Copying followed by even separation and cell division produces matching new body cells; more body cells support growth; new body cells can replace damaged cells during repair.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Aspire

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Predict what happens if copied chromosomes separate unevenly. Explain why a correct final split cannot fix the sets. [3]

One new nucleus may gain a copy while another lacks it; the sets no longer match; the final split does not redistribute the chromosomes between nuclei.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Explain why copying DNA after mitosis would break the sequence needed for matching new body cells. [3]

DNA copying must precede mitosis; each new nucleus needs one copy of every chromosome; copying after separation cannot supply the missing matching copies during that earlier separation.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Compare the two faulty models: copying skipped, and both A copies sent to one end. Repair each first error, then link the repaired sequence to new cells for repair. [4]

Copy DNA in interphase before mitosis; in anaphase send one copy of each chromosome type to each end; the later final split produces matching new body cells; these can replace damaged cells.

Maximum 4 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Choice

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Choose the order of interphase, mitosis and cytokinesis. Give one reason why copying comes first. [3]

Interphase before mitosis; cytokinesis last; DNA copying first supplies copies for both new nuclei.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Each new body cell needs A and B. Choose one copy of each or both A copies at one end. Explain your choice. [2]

One copy of each type to each end; each new nucleus then has a matching A and B set.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Choose new matching cells or uneven chromosome sets to help replace damaged skin cells. Use copy, separate and split to show why. [3]

New matching cells; DNA is copied and the copies separate into matching sets before the final split; new body cells can replace damaged cells.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Science staff guidance

Invite a response, give thinking time and allow a pass. No learner is required to read difficult text aloud. Accept point, sign, speech, drawing, exact scribing and adult-directed handling.

Tell the teacher the barrier observed: reading, vocabulary, processing, regulation, writing or science. Avoid prompting through every question or supplying the final answer.

First check: DNA is copied before mitosis. Second check: both new nuclei receive a matching set. Third check: the cell still needs the final split. Re-model only the insecure step, then use a fresh phase or number case.

The teacher engages with an actual review response and uses the exit to change the next model or prompt. Use one usual-mode response as evidence. Keep the individual support plan in charge of regulation and access.

Made by Matt

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TA Brief

At check-out, record E for justifying the cell-cycle order and explaining how matching chromosome sets support growth and repair against the pupil's target.

I am learning to justify the cell-cycle order and explain how matching chromosome sets support growth and repair.

Key checks: copy DNA in interphase; separate matching chromosome copies during mitosis; split the cell in cytokinesis. The four phases of mitosis are prophase, metaphase, anaphase and telophase.

Offer the full reference beside the sequence builder. Hear the reason for copying before mitosis, one copy of each chromosome per new nucleus, and the final split into separate cells. Wait, gesture, rephrase, then model one type before a fresh pupil move. Give time for quiet partner rehearsal before writing.

Hinge: six chromosomes in each daughter. Parallel check: eight in each daughter. Hear the reason and use the full answer guide for a misconception diagnosis.

At You do, record actual I · S · C beside the task. Access support and content prompts are recorded separately. Use a quiet pause and one-step return in line with the individual plan.

Report what the learner showed, the barrier, the support and the next move. A correct click alone is not an explanation.

Feedback codes VF · WS · I · NS + … · E · R · // · ?
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Stage timings

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I do	3
We do	7
I do	2
We do	5
You do	8
Review and improve	4
Exit ticket	3
Tools & print	0

Total: 40 minutes

Made by Matt